Primary n=13 IC toy-curve certificate

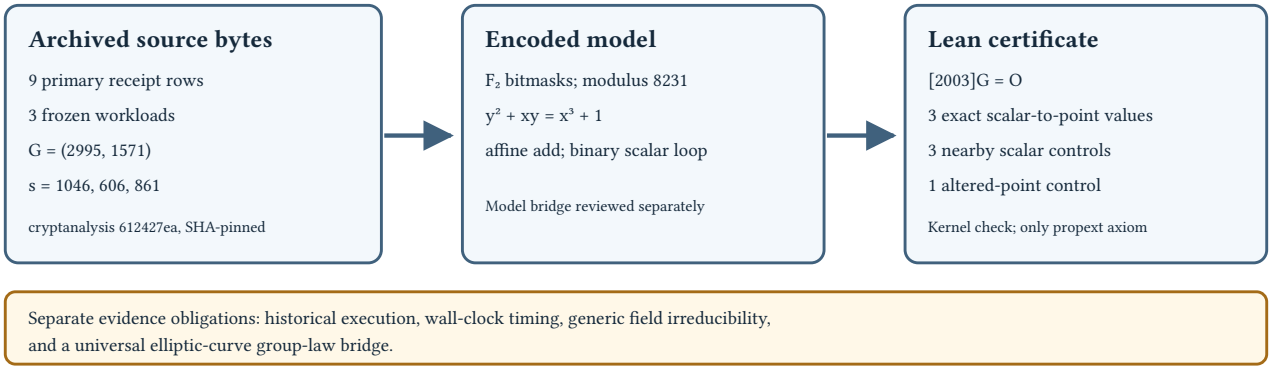
Technical review · 2026-11-07 UTC · Claim IC-PRIMARY-N13-ENCODED-SCALARS-20261107

The exact Lean 4.35.0 declaration checks three archived scalar-to-point equalities in a separately defined 13-bit arithmetic model. The source-fidelity review and independent arithmetic calculation agree with those finite values.

Scalar	Computed encoded point	Workload
1046	(2659, 1282)	`026858681e66`
606	(6749, 6034)	`398002869ca3`
861	(4412, 6211)	`aa4b90819f2f`

The theorem also checks the modulus mask 8231, generator curve membership, `scale 2003 G = none`, three adjacent-scalar inequalities, and an altered coordinate that fails the encoded curve equation. The exact direct Lean run exited 0; axiom printing for the certificate and its helpers lists only the standard Lean axiom `propext`. The hosted full workspace gate remains a merge requirement.

Primary n=13 toy-curve proof boundary



Source: cryptanalysis commit `612427ead3d47565b92da3e663739d5c17ce71ed`, archived primary JSONL SHA-256 `3886d486bf05ba476c00a67c9d7b20ed07250862273019a3b59195a22d60b973`. Proof-input commit `83d750c7a0`; exact source-to-theorem mapping and workload hashes are in `source.md` and `reviews/fidelity.md`.

Review and claim boundary

The independent source-fidelity report found the statement faithful to the fixed encoded computation. The independently coded algebra report reproduced the points and controls; it also checked all nonzero field encodings against the inversion exponent. The first algebra reviewer read more of the original Markdown plan than its intended allowance, including a favorable prior. `blind_addendum.md` records that protocol deviation and freezes a new input without target outputs for a further result-blind computation. That third review independently computed all ten requested multiples by repeated addition and binary double-and-add; after its report was frozen, comparison found agreement on all three targets and adjacent controls. The altered point residual was 2658. Its file is `reviews/strict-blind.md`, SHA-256 `3e64d272a1106128872ec657e6b5350223fb00a18e39cc2295f3bc717c5a3833`. Conductor T-1868 later displayed `cancelled` after claim release; its frozen report remains the technical record.

The precommitted plan and reports are Markdown. The repository independence checker expects YAML plan and attestation blocks with `TASK-*` IDs, whereas the actual Conductor IDs are `T-*`; its run returned exit 2 with `no review_plan block`. This round retains the original files and read lists for manual inspection and does not claim a passing machine independence check.

Verified finite statement: fixed encoded arithmetic equalities and negative controls, with pinned Lean compilation, source-fidelity review and a result-blind independent computation. Machine status is `passed` locally; fidelity is `faithful_scoped`; review outcome is `validated_scoped` for the finite computation. Hosted workspace CI is still a publication gate.

Separate obligations: a Lean proof of field irreducibility and generic point group law, exact order, equivalence to the cryptanalysis implementation, and independent authentication of historical execution and timing. None follows from the finite certificate.

Detailed commands, hashes, receipts and review attestations are in `machine-check.json`, `source.md`, `review_plan.md`, `blind_addendum.md`, and `reviews/` beside this PDF.